

AI 모델 성능 개선 결과

결론

- 관심수위 + MAE + 2020년 이후 학습 이 그나마 안정적인 성능
- 22.07.12 유형 사상 : 궁내교 , 대곡교 두 지점 동시에 성능이 저하
→ (1) 입력 강우/유량/수위 시계열 정합 문제, (2) 운영(방류/펌프/수문) 개입, (3) 센서 이상/결측/시간오프셋 중 하나(또는 복합) 가능성

12시간 예측

72시간 예측 후 뒷부분 날리고 앞 18시간만 사용

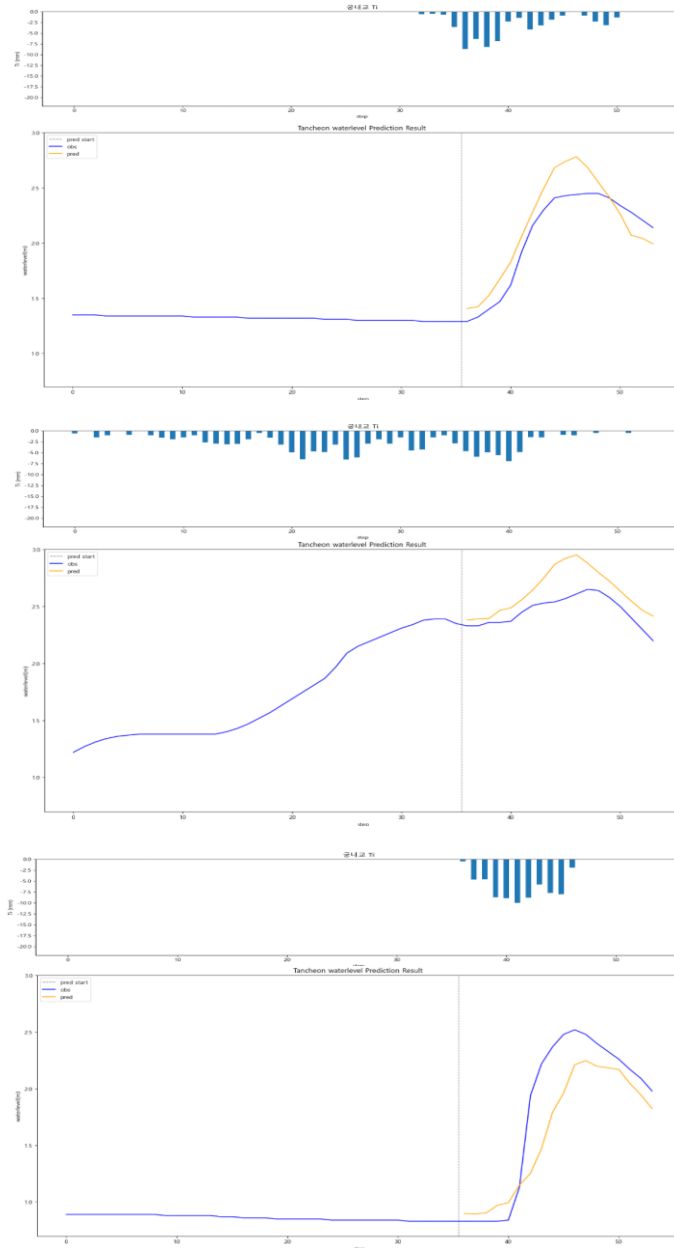
12시간 예측 (中 앞 3시간)

- » 사상별 편차가 매우 큼
- » 특히 22년 사상 R^2 음수

궁내교					대곡교			
Test data	r2	mse	mae	rmse	r2	mse	mae	rmse
20.08.02	0.7961	0.0359	0.1706	0.1896	0.4887	0.1246	0.3344	0.3531
22.07.12	-1.3191	0.0375	0.1701	0.1938	-1.7001	0.0381	0.1793	0.1953
23.07.11	0.7596	0.1095	0.2473	0.3309	0.8885	0.1440	0.3038	0.3794
24.07.23	0.4878	0.0606	0.1629	0.2461	0.2750	0.0411	0.1598	0.2028
25.07.16	0.4308	0.0383	0.1521	0.1958	0.8926	0.0597	0.2114	0.2443
25.08.14	0.2319	0.0236	0.1327	0.1539	0.8466	0.0373	0.1475	0.1932

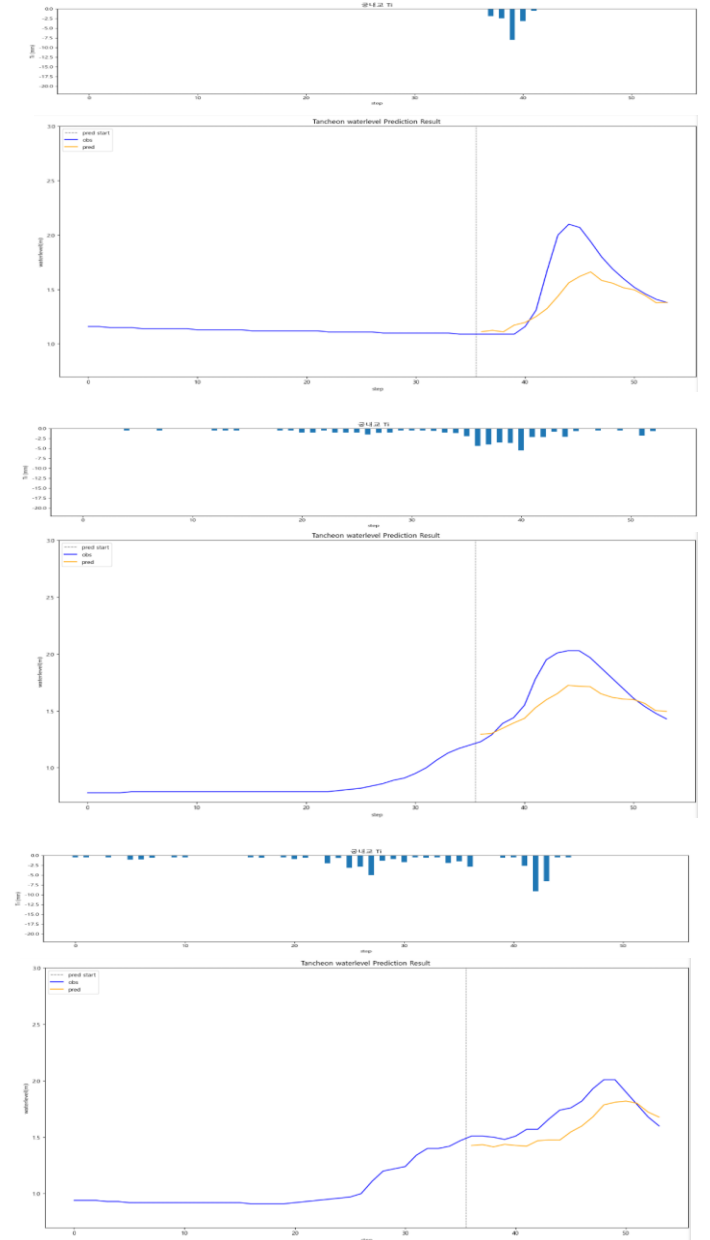
testdata_200802

R2 : 0.7961



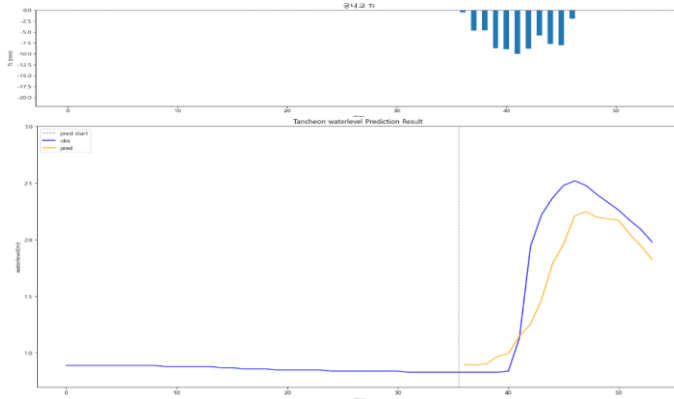
testdata_240723

R2 : 0.4878



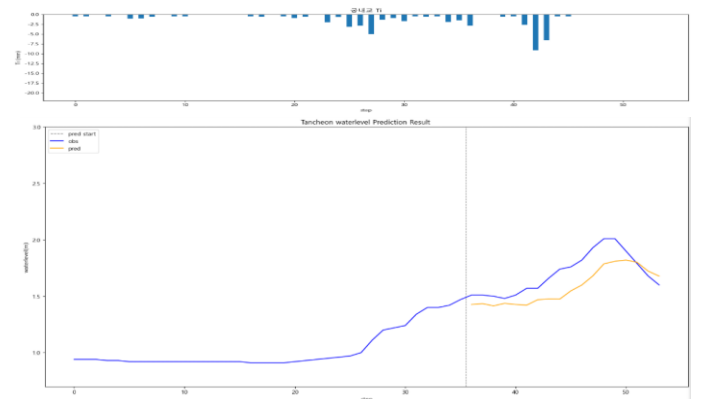
testdata_220712

R2 : -1.3191



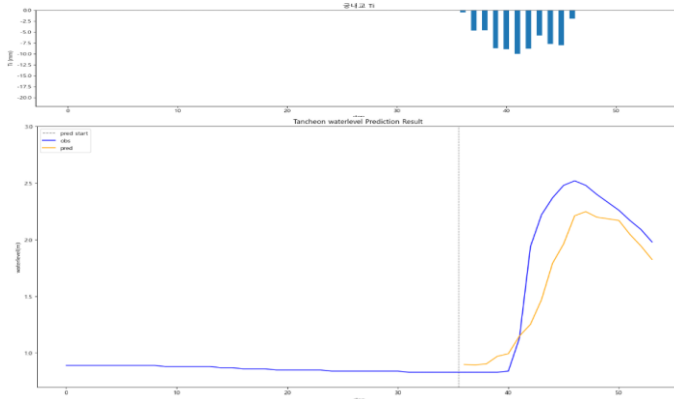
testdata_250716

R2 : -2.8628



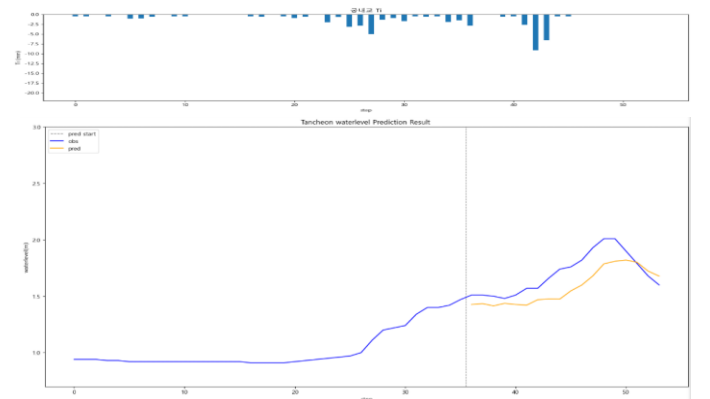
testdata_230711

R2 : 0.7596



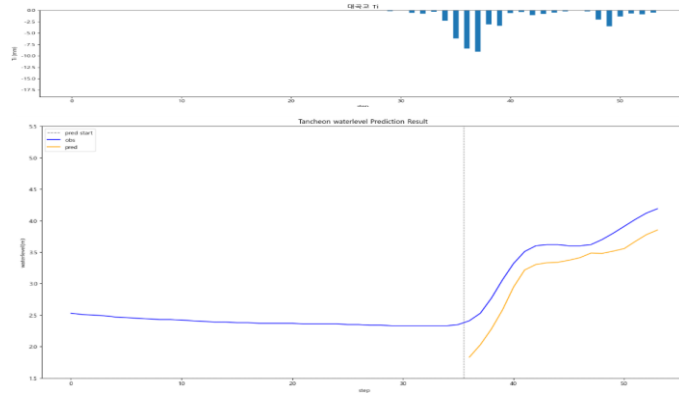
testdata_250814

R2 : 0.2319



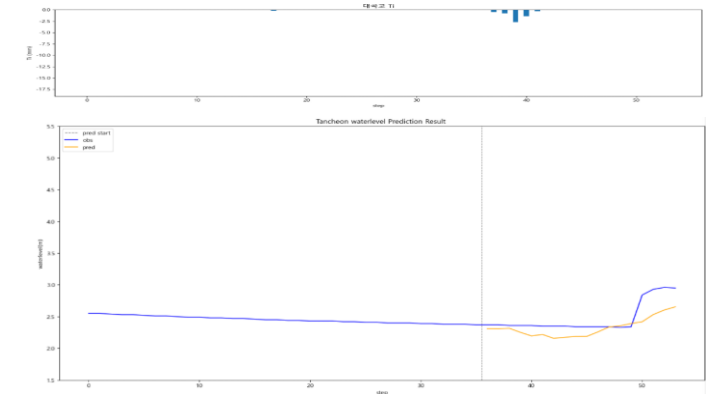
testdata_200802

R2 : 0.4887



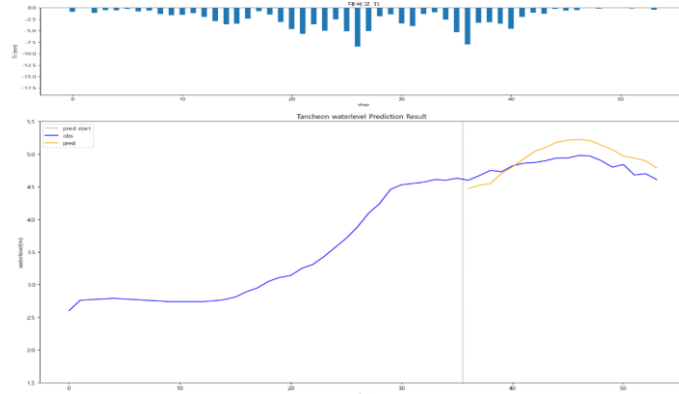
testdata_240723

R2 : 0.2750



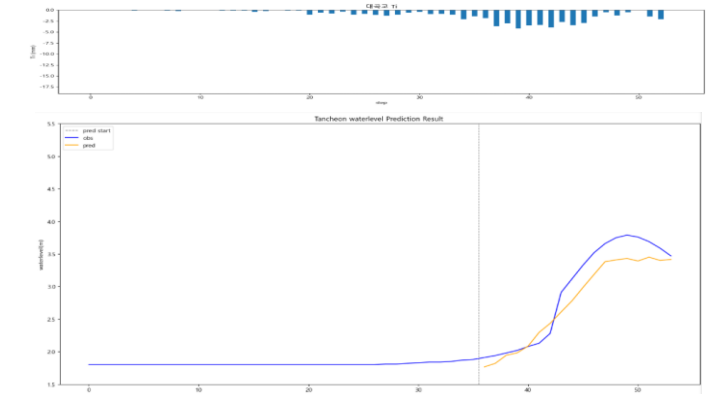
testdata_220712

R2 : -1.7001



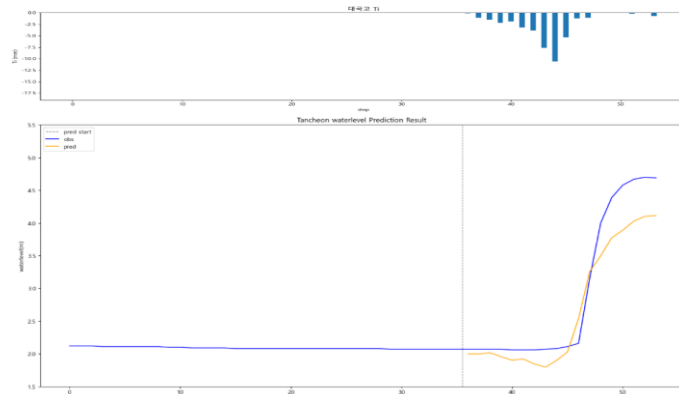
testdata_250716

R2 : 0.8926



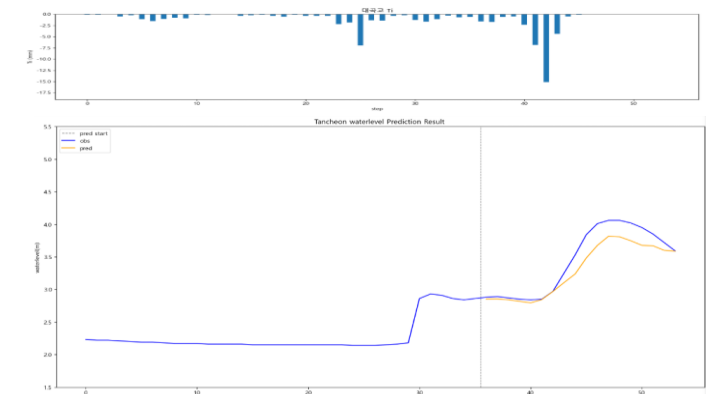
testdata_230711

R2 : 0.8885



testdata_250814

R2 : 0.8466



**관심수위 + MAE +
2020년 이후 데이터 학습**

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)
- smoothing : 궁내교 (1) 대곡교 (2)
- 그래프 스케일 고정 : 궁내교 (0.7 - 3) 대곡교 (1.5 - 5.5)

-
- learning_rate = 0.0005
 - Loss_function = MAE
 - Seq_length = 36 (6시간)
 - Num_of_layer = 18

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 , 검증 : 총 40개 中 35개 데이터 파일 사용

테스트 : 학습 데이터에서 제외한 5개 데이터 파일 (4개 이벤트)

+ 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : 2020 ~ 20240920 (25개)
- 검증 : 20240921 ~ 250919 85번 (7개)
- 테스트 : 250919 86번 ~ 251006 (3개)

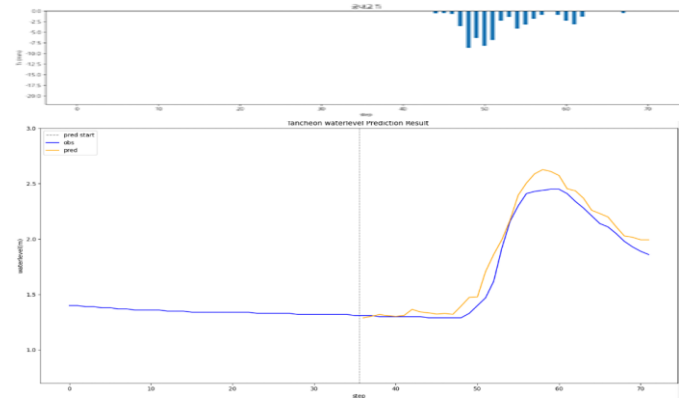
관심수위 + MAE + 2020년 이후 데이터 학습

- » 궁내교 : 최근 패턴 집중 → 소폭 개선
- » 대곡교 : 학습 데이터 감소 → 일반화 성능 저하
- » 지점별 데이터 분리 ???

궁내교					대곡교			
Test data	r2	mse	mae	rmse	r2	mse	mae	rmse
20.08.02	0.9513	0.0102	0.0814	0.1013	0.8845	0.0704	0.1797	0.2653
22.07.12	-0.8548	0.0735	0.2146	0.2712	-0.1561	0.1619	0.3550	0.4023
23.07.11	0.8751	0.0349	0.1217	0.1868	0.7304	0.2453	0.3701	0.4953
24.07.23	0.8136	0.0150	0.0610	0.1227	0.6077	0.0186	0.0981	0.1365
25.07.16	0.8213	0.0179	0.0927	0.1338	0.8647	0.0624	0.1862	0.2498
25.08.14	0.7835	0.0152	0.0984	0.1235	0.8840	0.0402	0.1572	0.2006

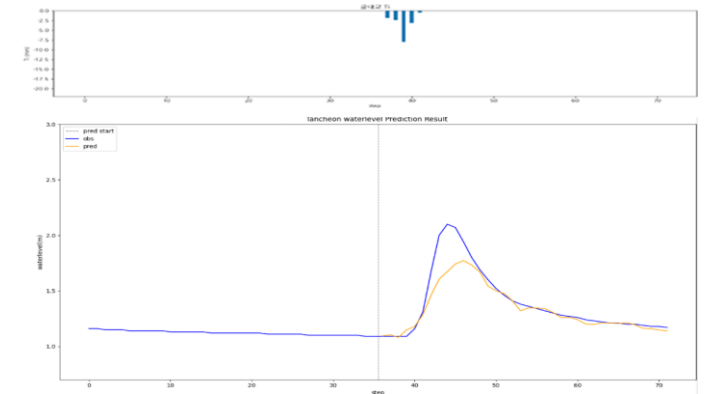
testdata_200802

R2 : 0.9513



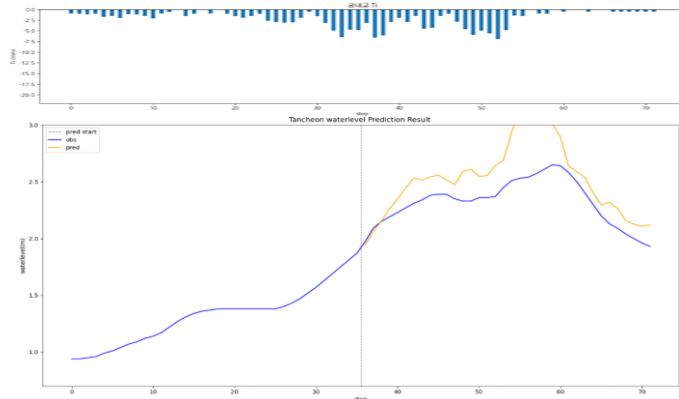
testdata_240723

R2 : 0.8136



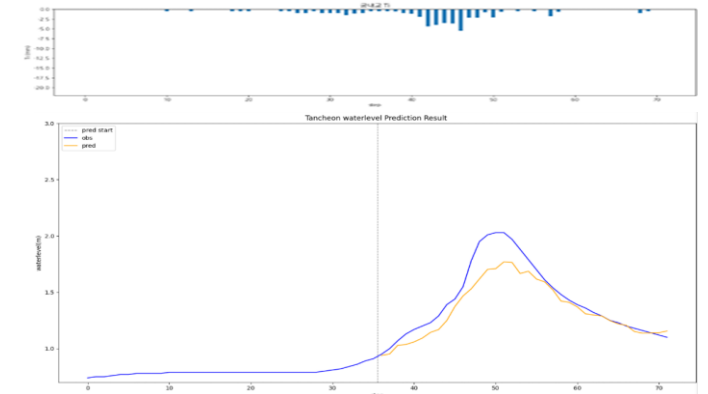
testdata_220712

R2 : -0.8548



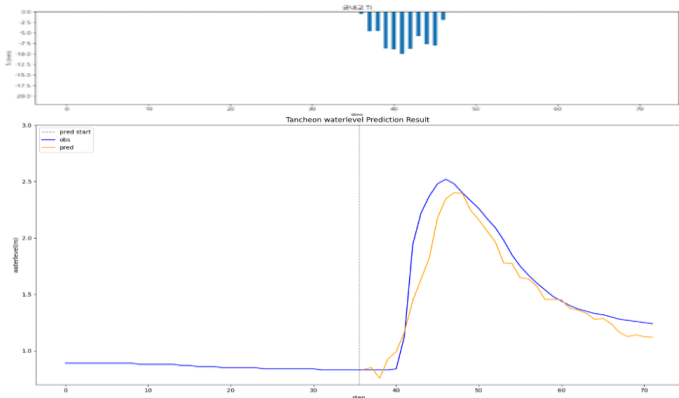
testdata_250716

R2 : 0.8213



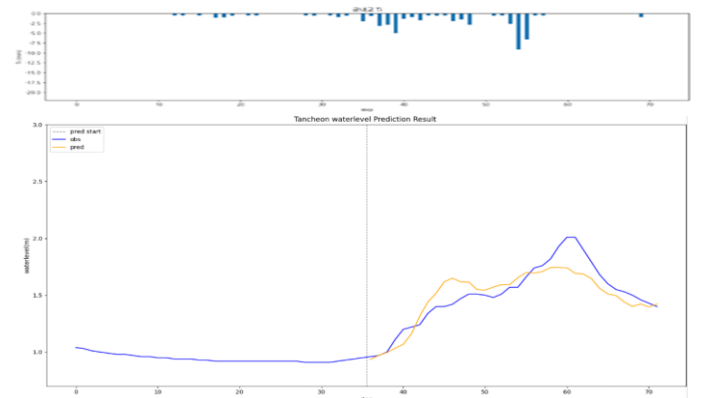
testdata_230711

R2 : 0.8751



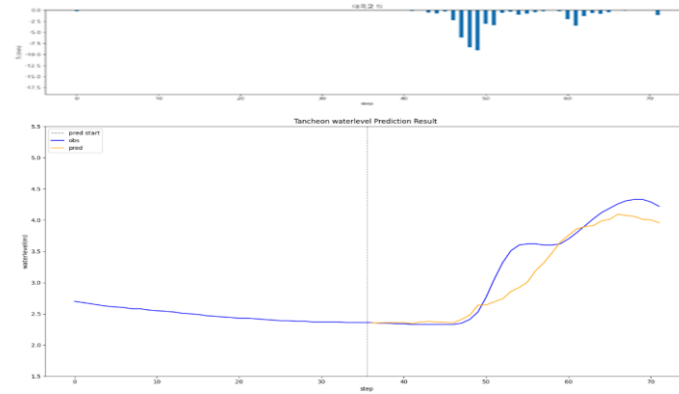
testdata_250814

R2 : 0.7835



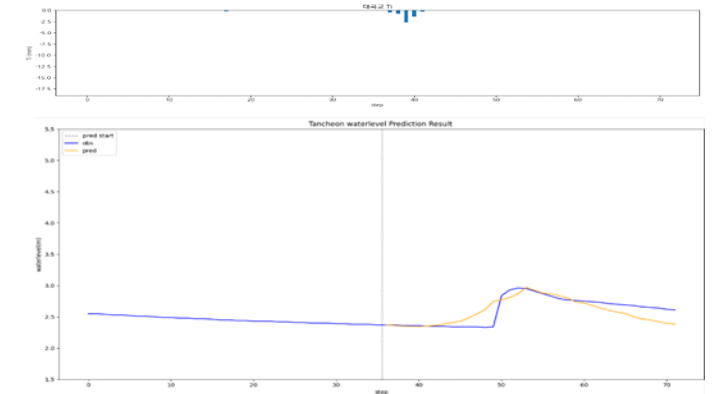
testdata_200802

R2 : 0.8845



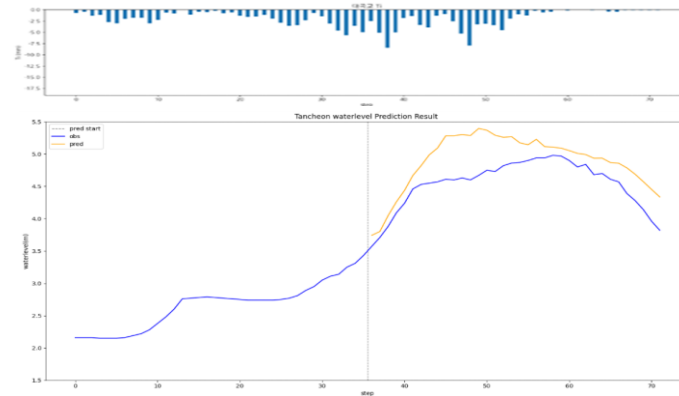
testdata_240723

R2 : 0.6077



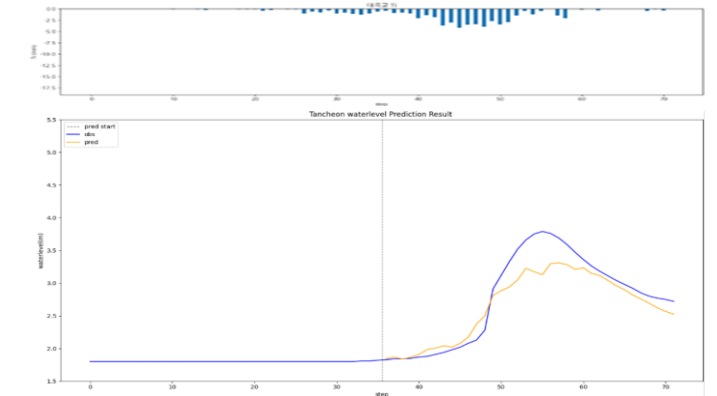
testdata_220712

R2 : -0.1561



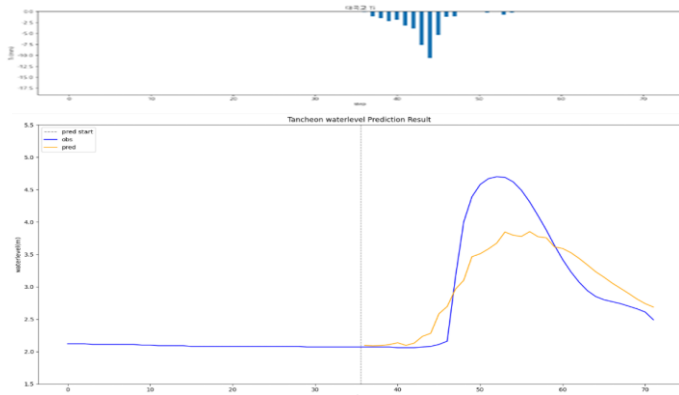
testdata_250716

R2 : 0.8647



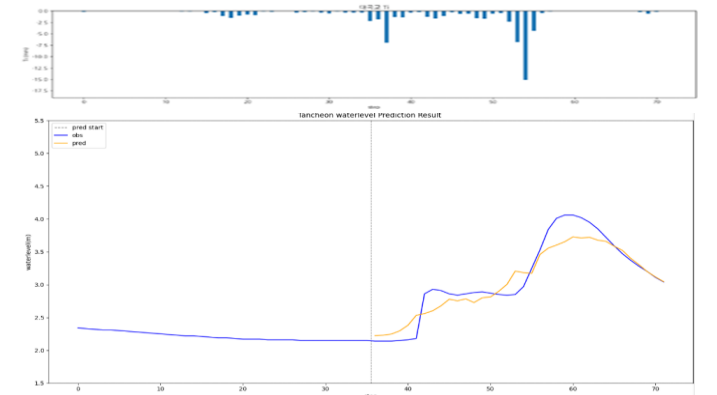
testdata_230711

R2 : 0.7304



testdata_250814

R2 : 0.8840



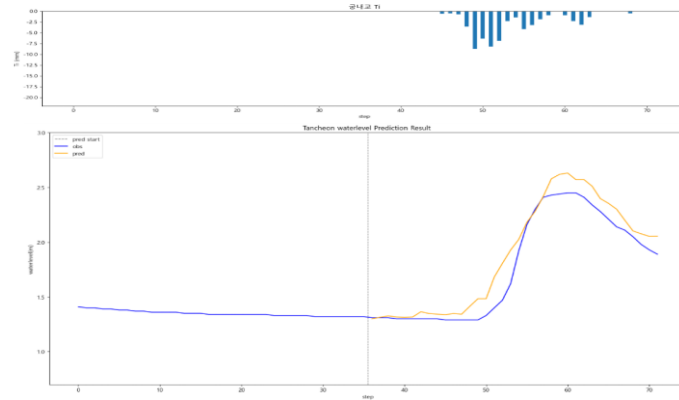
**관심수위 + MAE +
2020년 이후 데이터 학습
10분 단위 예측**

궁내교 - 10분단위

* model_(gn/dg)_20

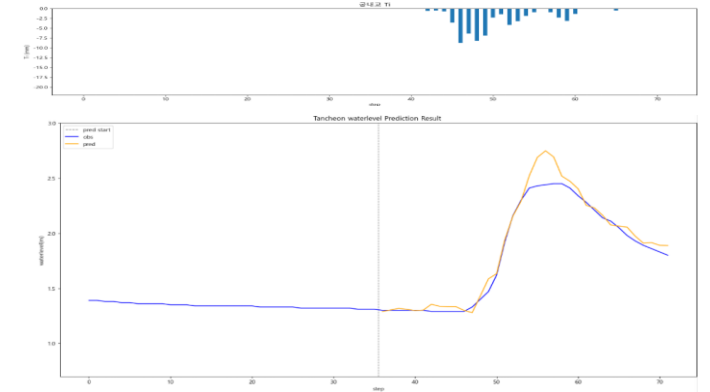
10분 전

R2 : 0.9138



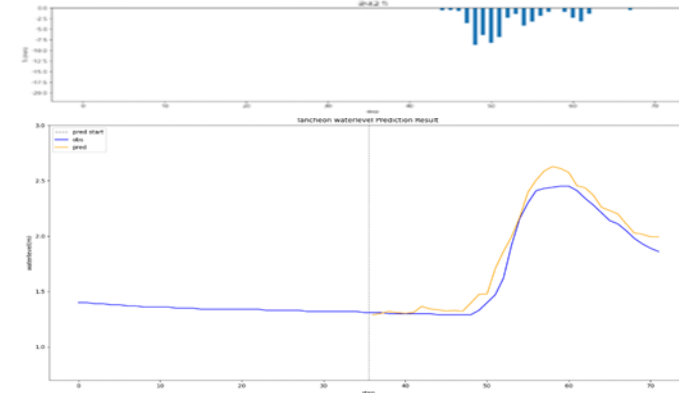
20분 후

R2 : 0.9579



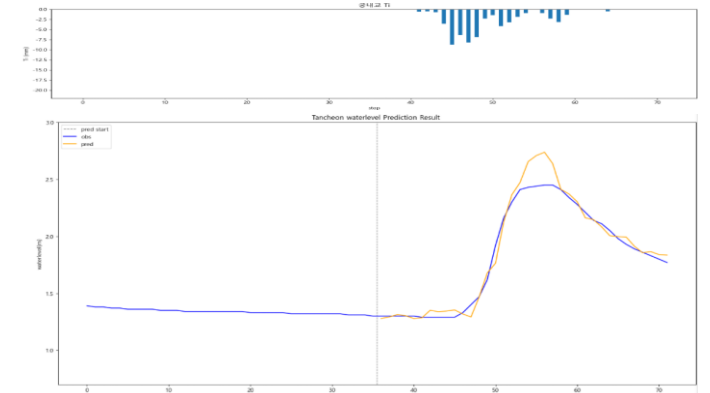
testdata_ 200802 (02)

R2 : 0.9513



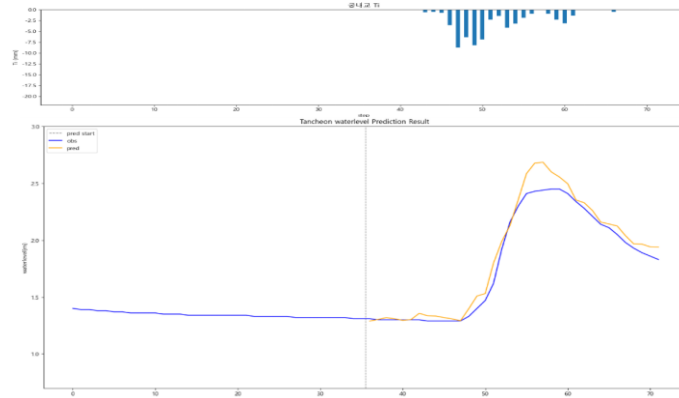
30분 후

R2 : 0.9529



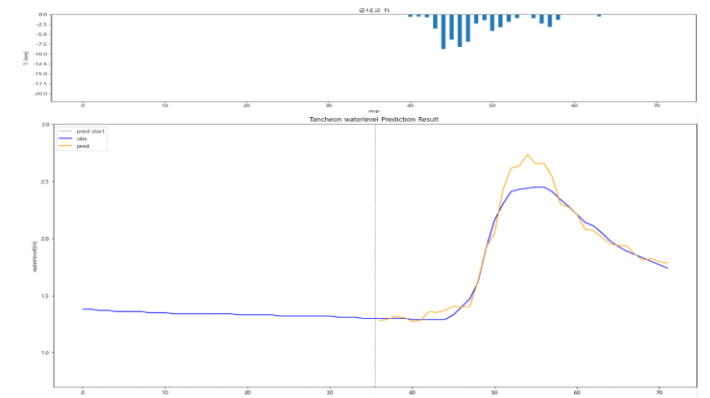
10분 후

R2 : 0.9585



40분 후

R2 : 0.9478

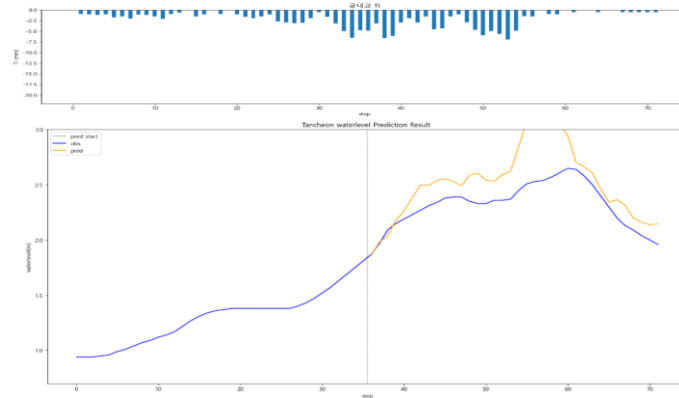


궁내교 - 10분단위

* model_(gn/dg)_20

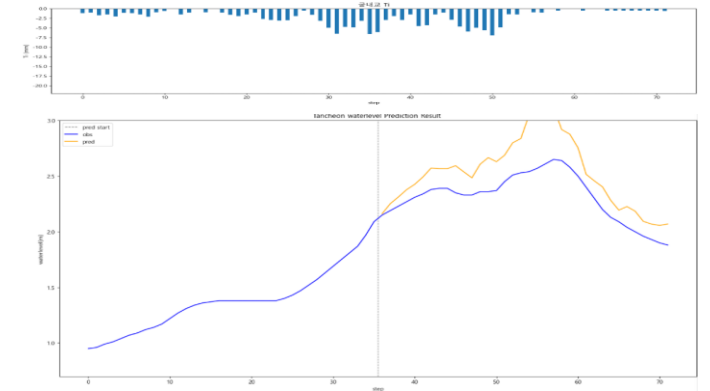
10분 전

R2 : -0.7405



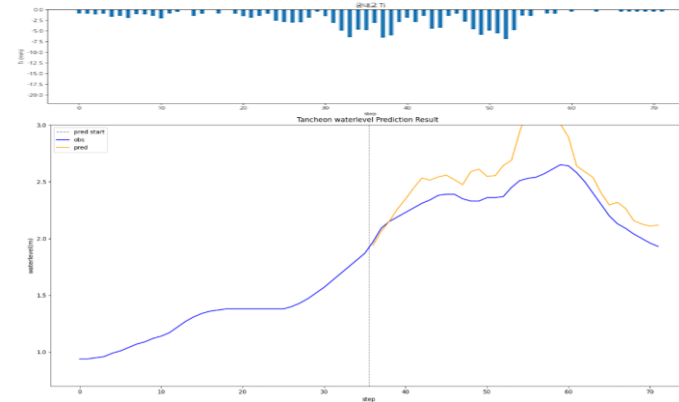
20분 후

R2 : -0.4821



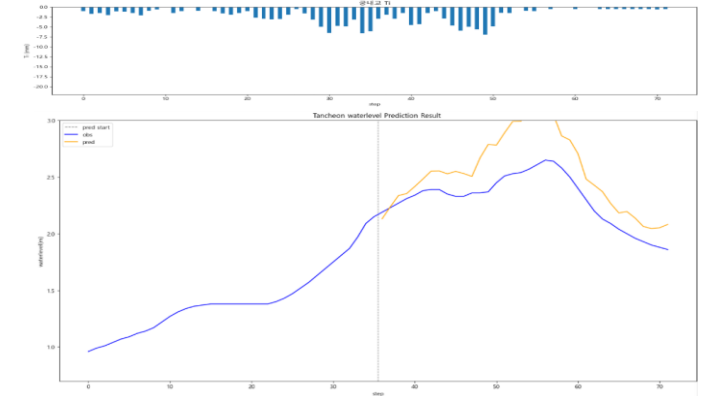
testdata_ 220712 (16)

R2 : -0.8548



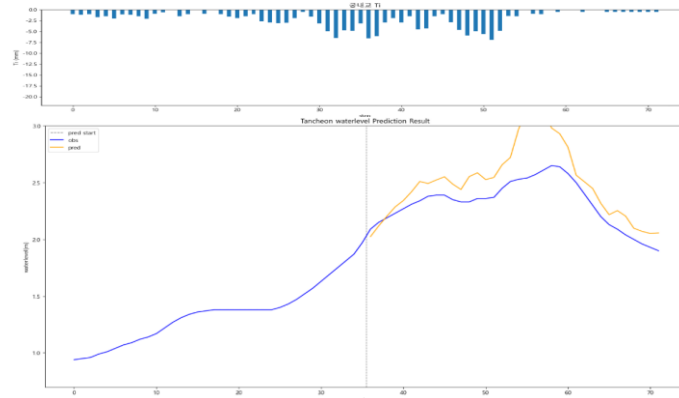
30분 후

R2 : -0.5463



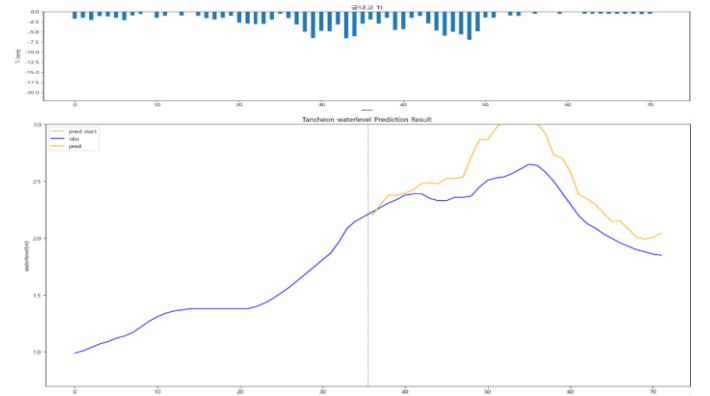
10분 후

R2 : -0.4408



40분 후

R2 : -0.2182

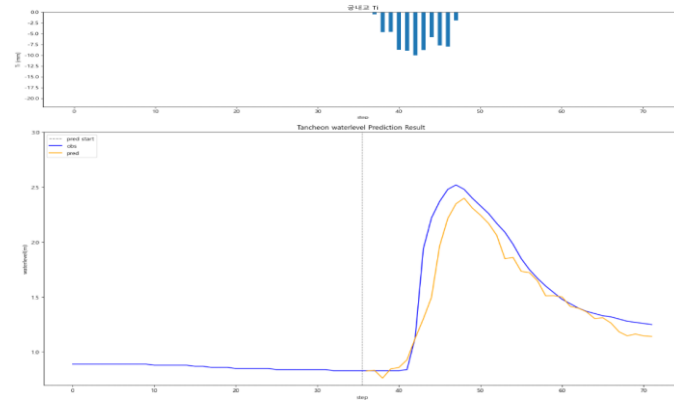


궁내교 - 10분단위

* model_(gn/dg)_20

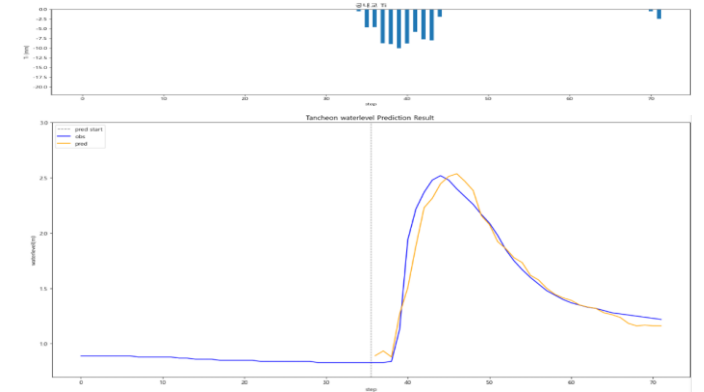
10분 전

R2 : 0.8657



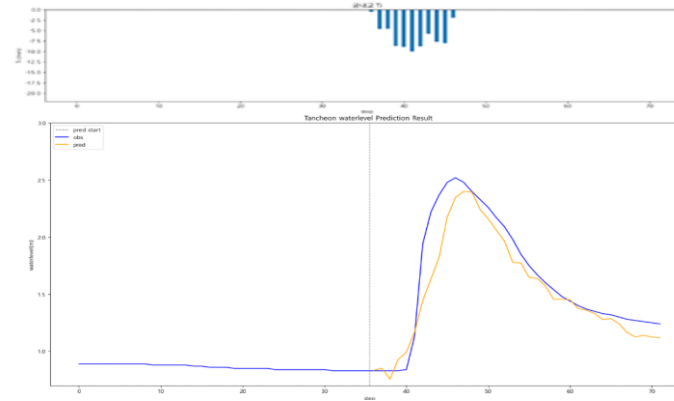
20분 후

R2 : 0.9471



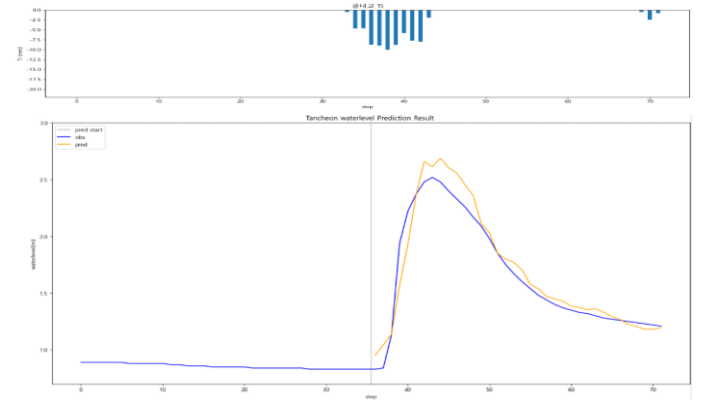
testdata_230711 (07)

R2 : 0.8751



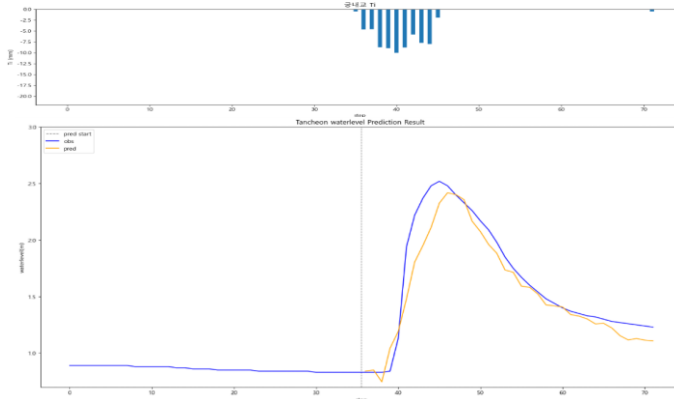
30분 후

R2 : 0.9322



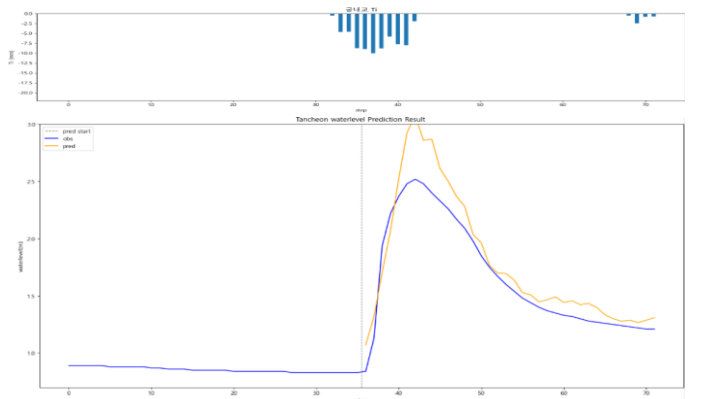
10분 후

R2 : 0.9012



40분 후

R2 : 0.8200

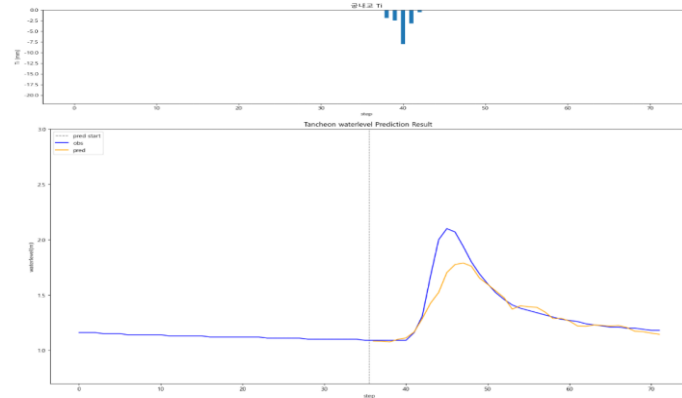


궁내교 - 10분단위

* model_(gn/dg)_20

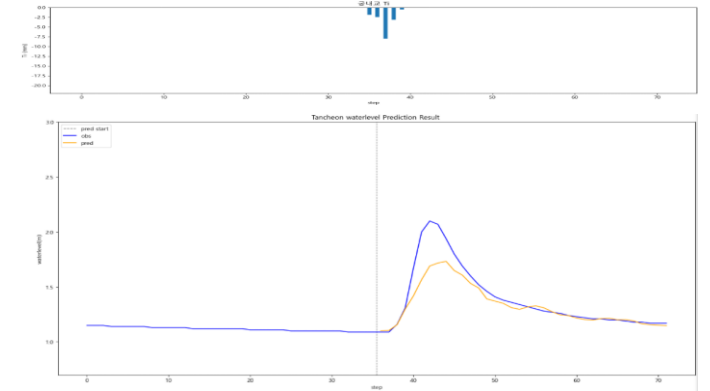
10분 전

R2 : 0.8064



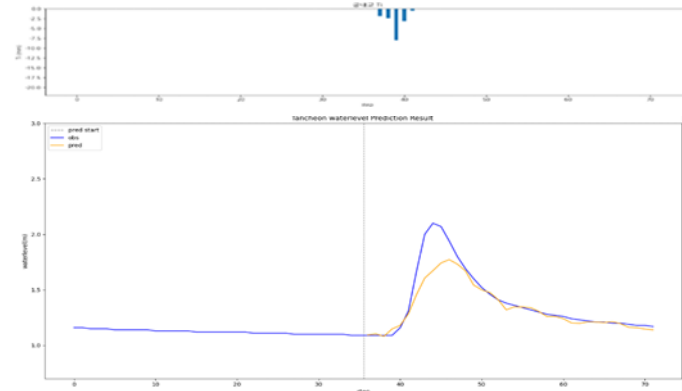
20분 후

R2 : 0.7748



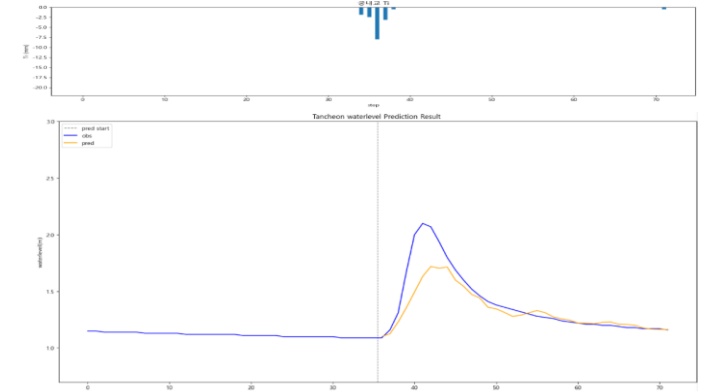
testdata_ 240723 (18)

R2 : 0.8136



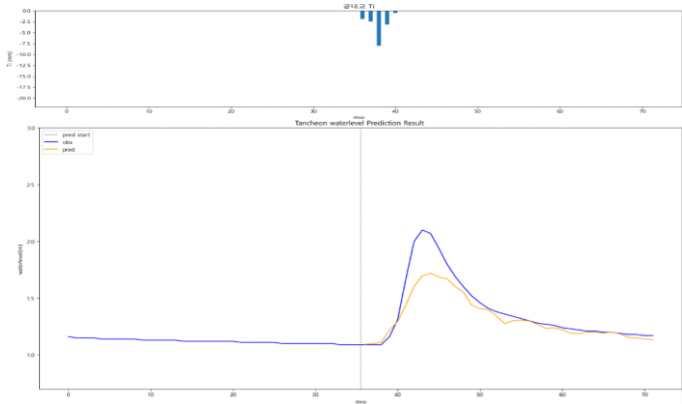
30분 후

R2 : 0.7132



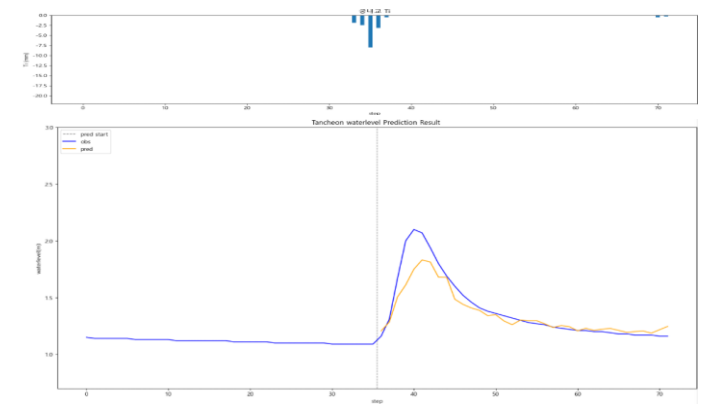
10분 후

R2 : 0.7844



40분 후

R2 : 0.8407

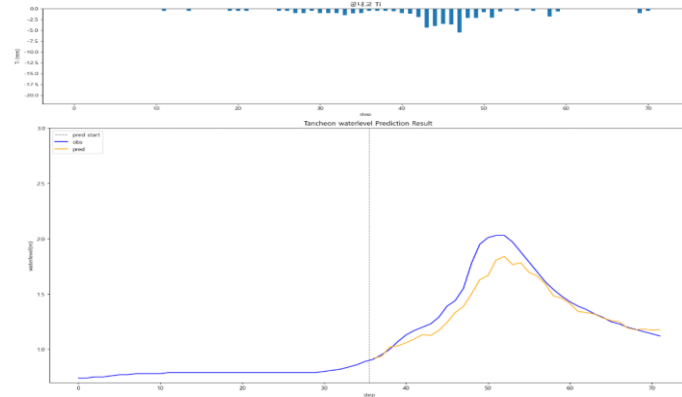


궁내교 - 10분단위

* model_(gn/dg)_20

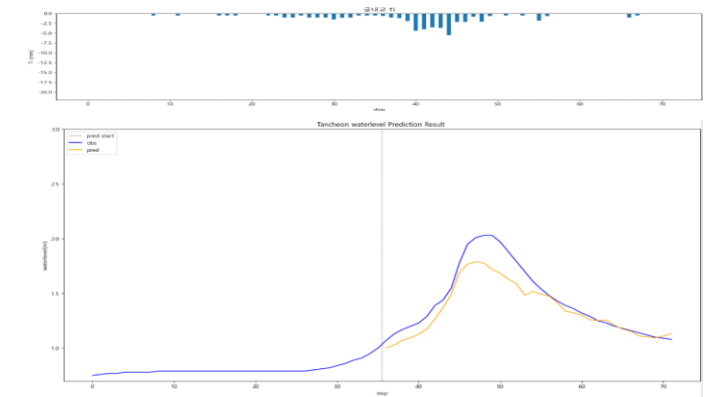
10분 전

R2 : 0.8512



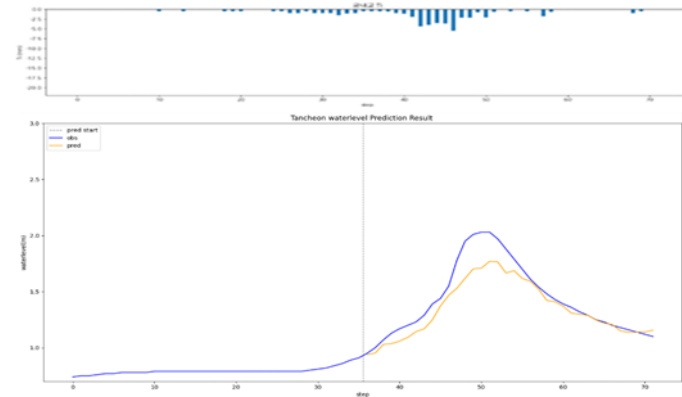
20분 후

R2 : 0.8289



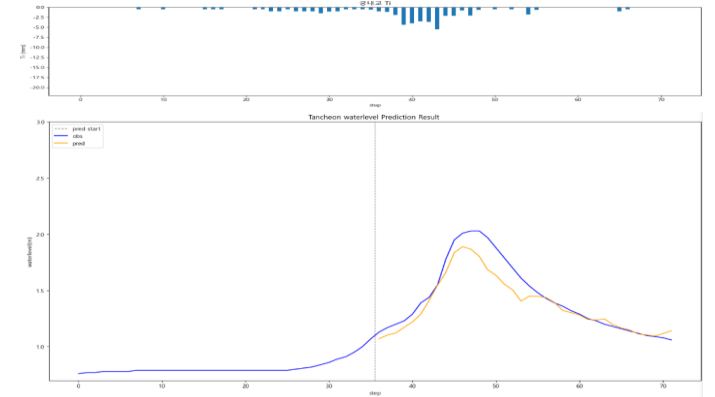
testdata_ 250716 (17)

R2 : 0.8213



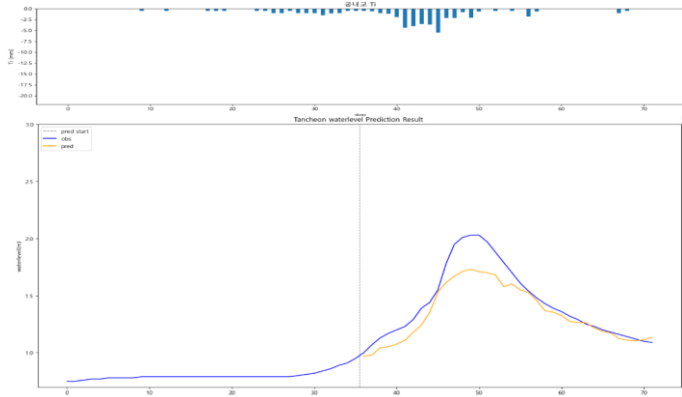
30분 후

R2 : 0.8696



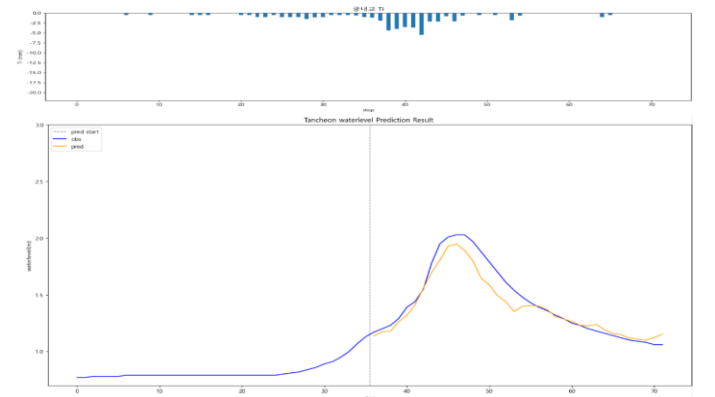
10분 후

R2 : 0.8073



40분 후

R2 : 0.9051

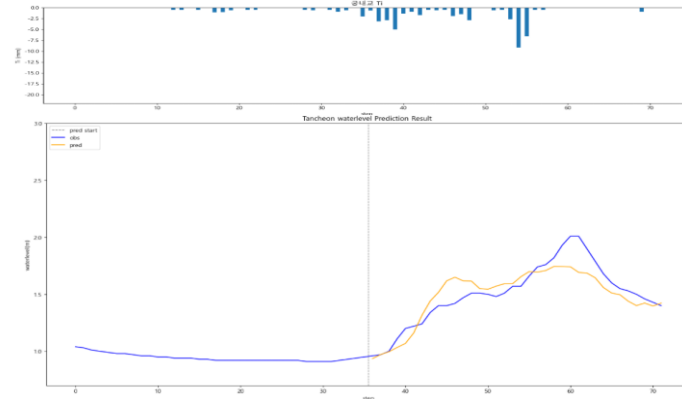
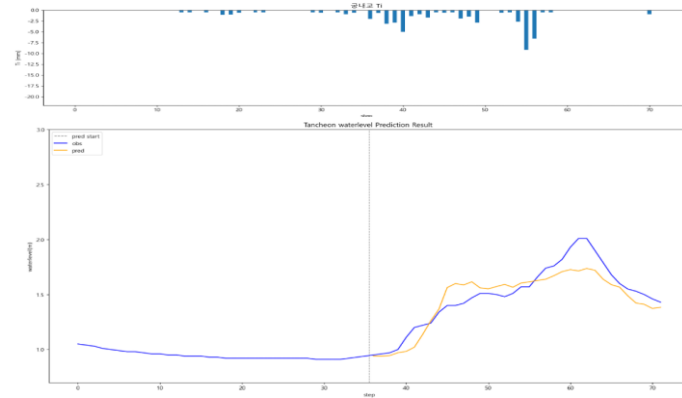


궁내교 - 10분단위

* model_(gn/dg)_20

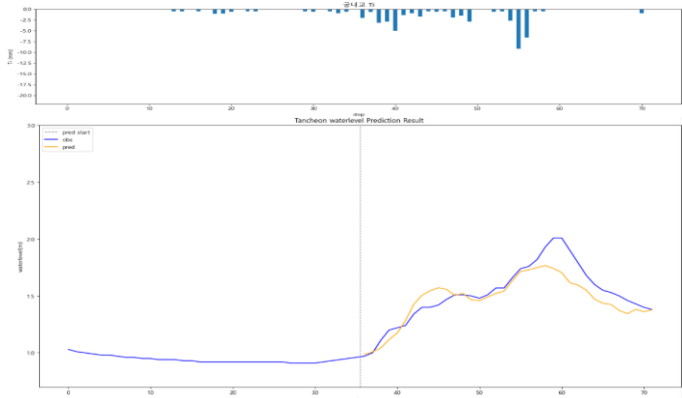
10분 전

R2 : 0.8109



testdata_250814 (05)

R2 : 0.7835

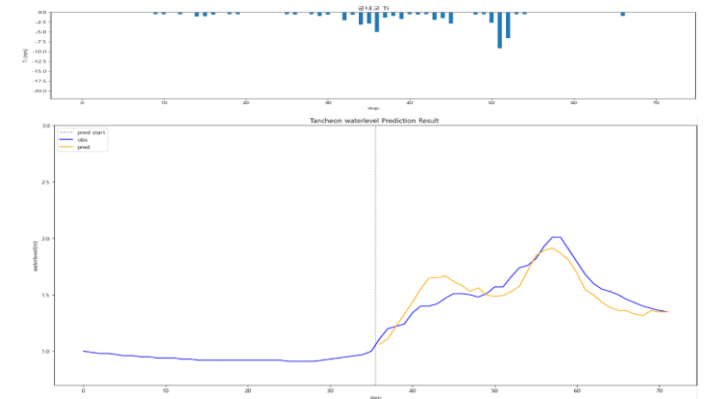
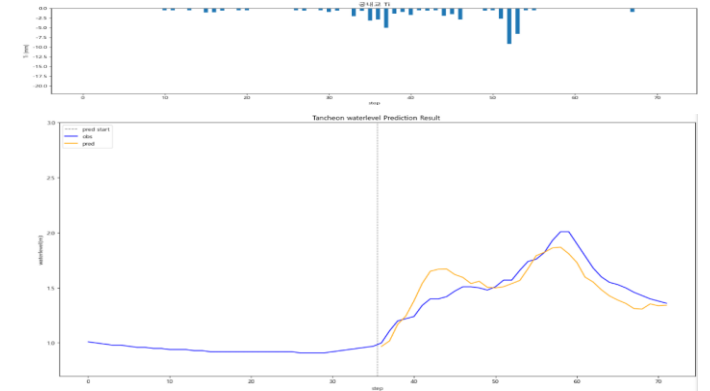


10분 후

R2 : 0.7776

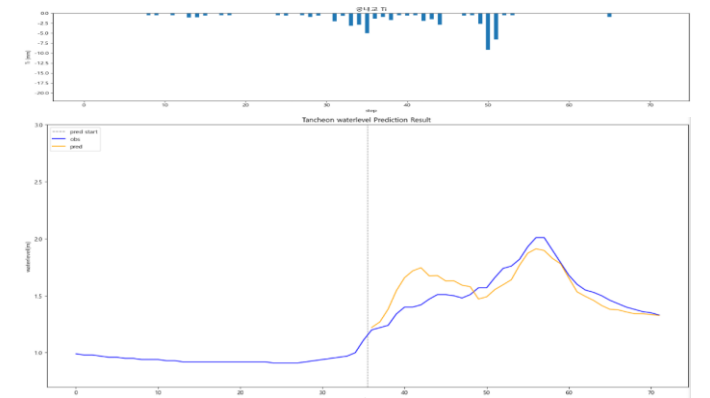
20분 후

R2 : 0.7068



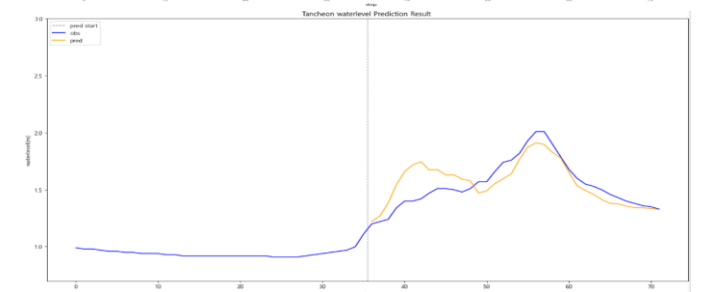
30분 후

R2 : 0.7407



40분 후

R2 : 0.6295

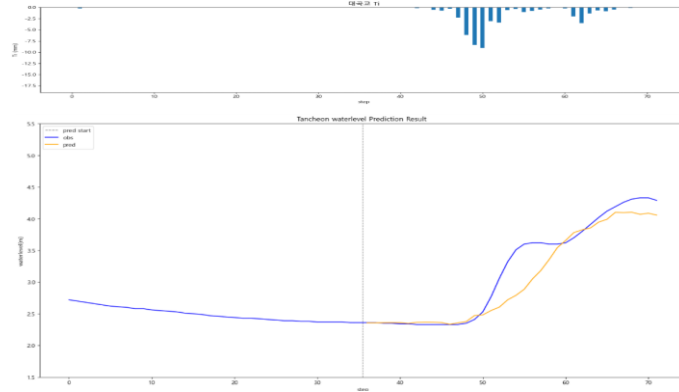


대곡교 - 10분 단위

* model_(gn/dg)_20

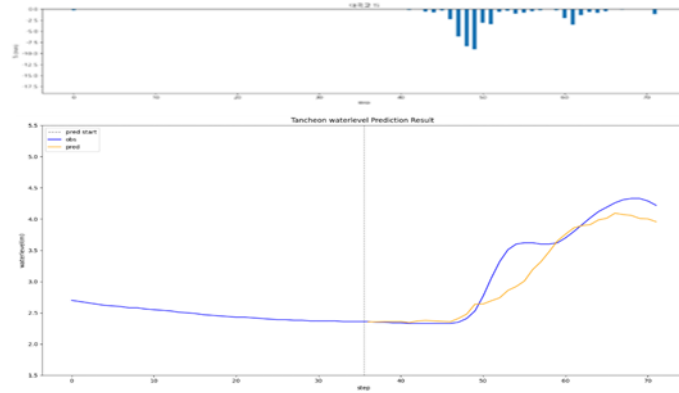
10분 전

R2 : 0.8834



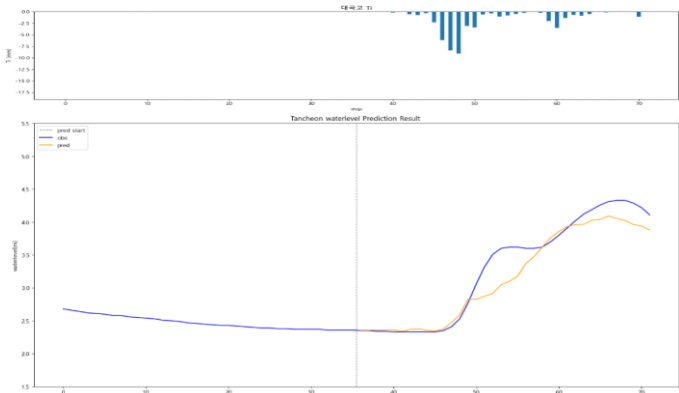
testdata_ 200802 (02)

R2 : 0.8845



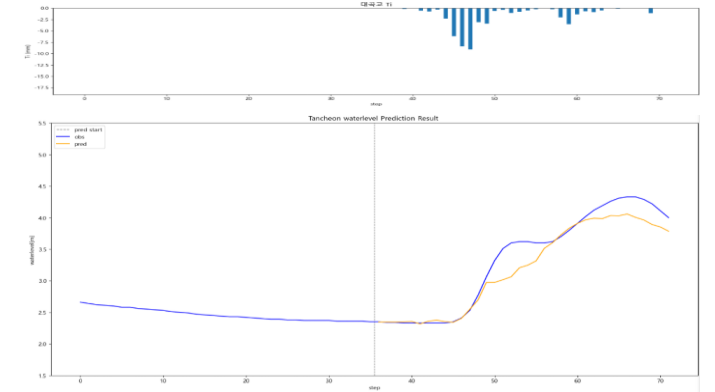
10분 후

R2 : 0.9069



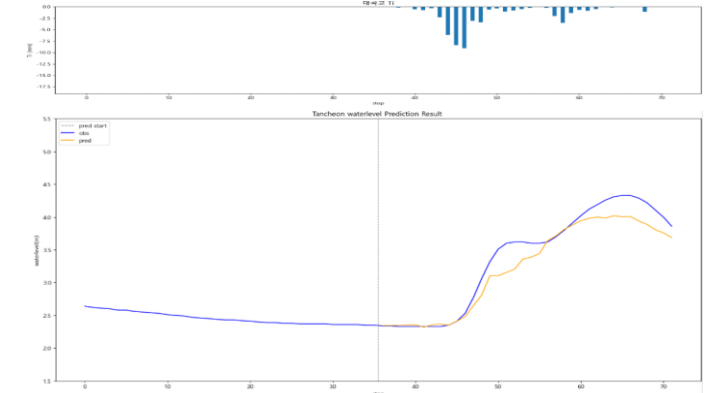
20분 후

R2 : 0.9173



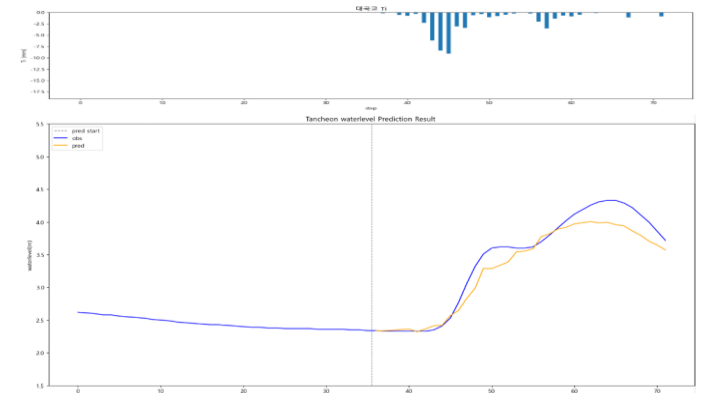
30분 후

R2 : 0.9201



40분 후

R2 : 0.9249

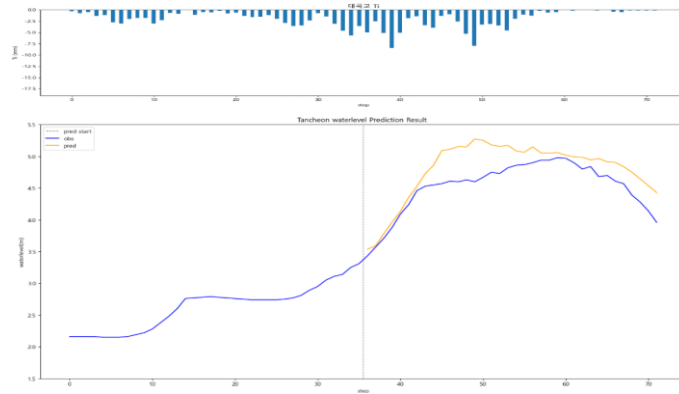


대곡교 - 10분 단위

* model_(gn/dg)_20

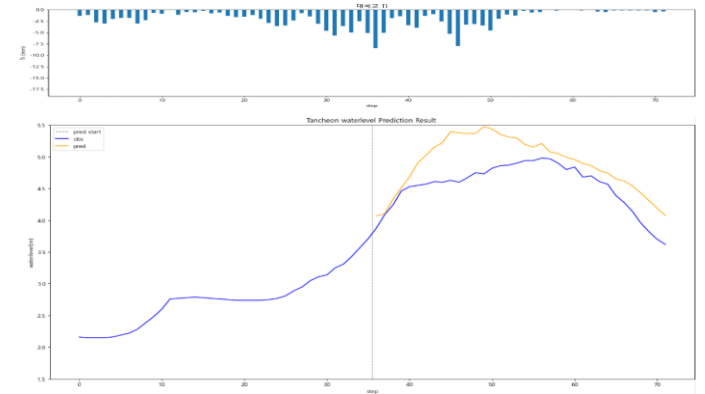
10분 전

R2 : 0.3503



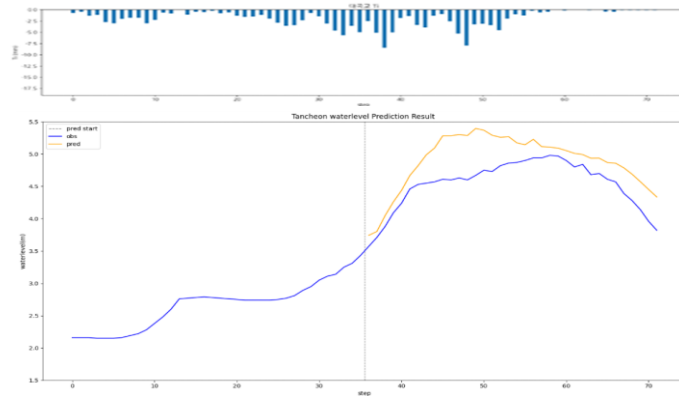
20분 후

R2 : -0.2738



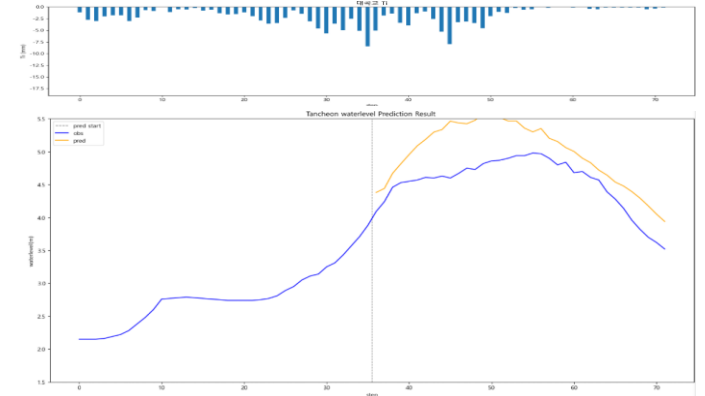
testdata_220712 (16)

R2 : -0.1561



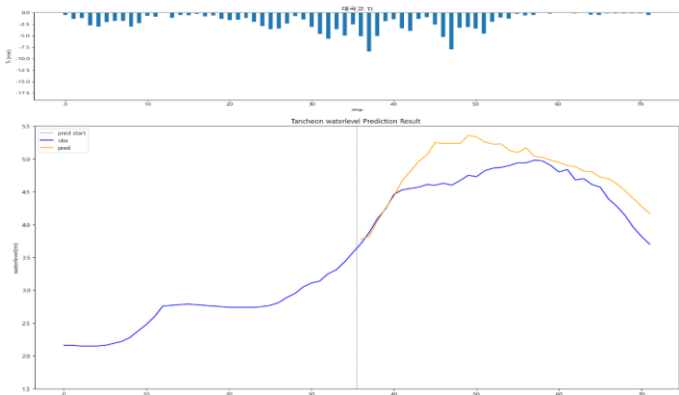
30분 후

R2 : -0.5470



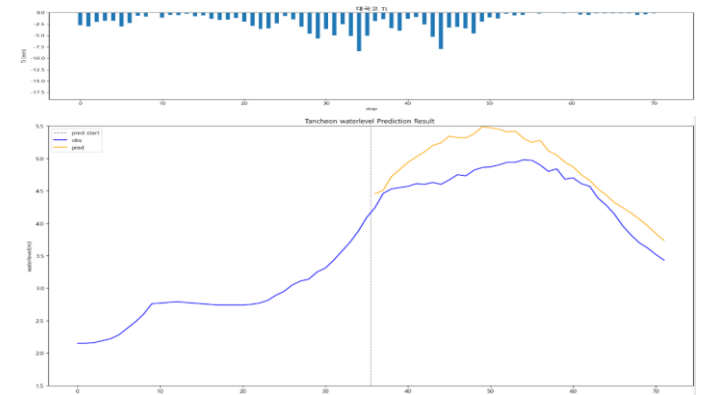
10분 후

R2 : 0.0810



40분 후

R2 : 0.1284

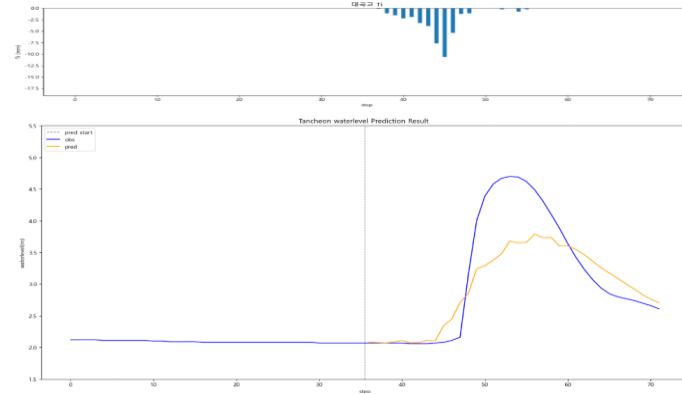


대곡교 - 10분 단위

* model_(gn/dg)_20

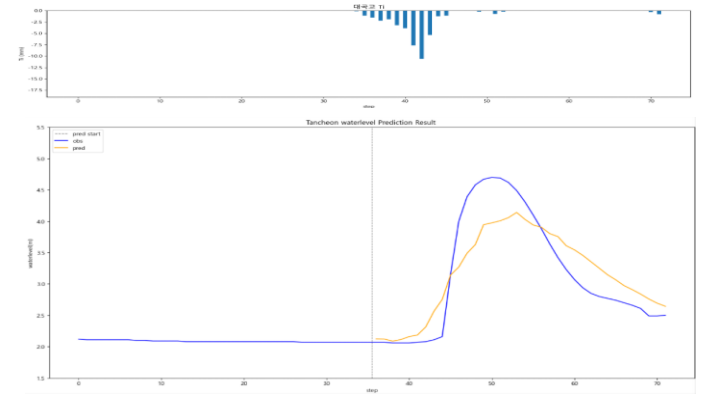
10분 전

R2 : 0.7046



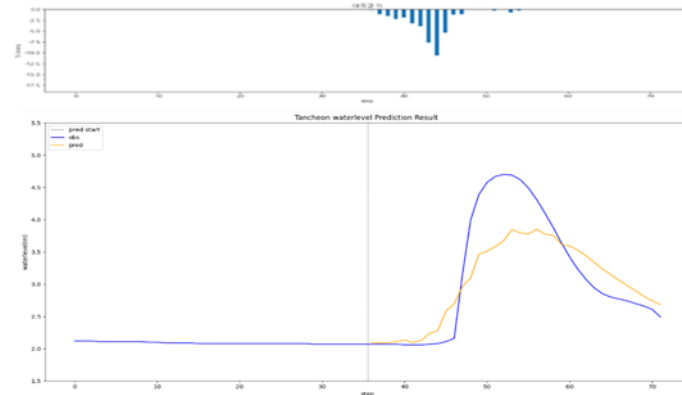
20분 후

R2 : 0.7840



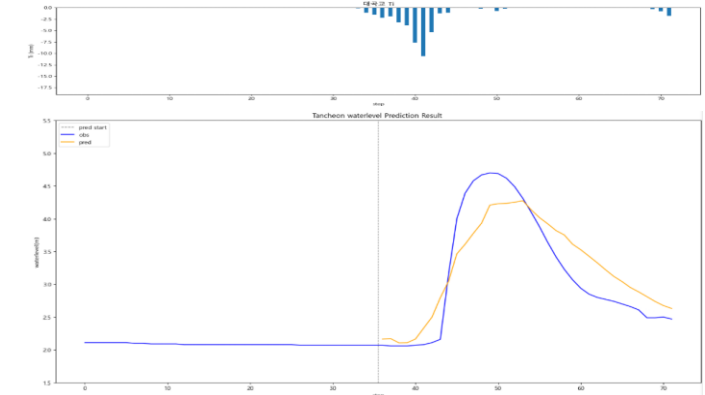
testdata_230711 (07)

R2 : 0.7304



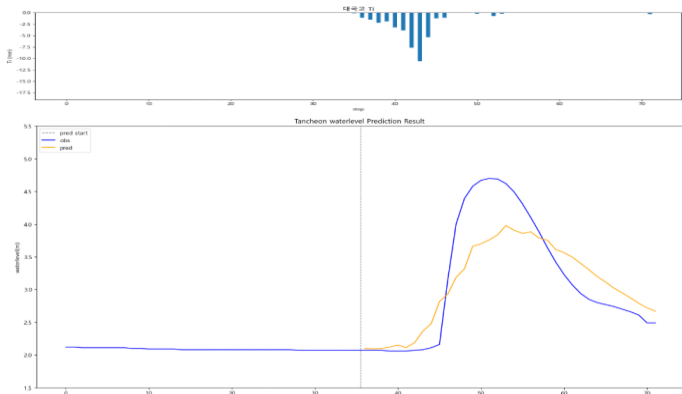
30분 후

R2 : 0.8028



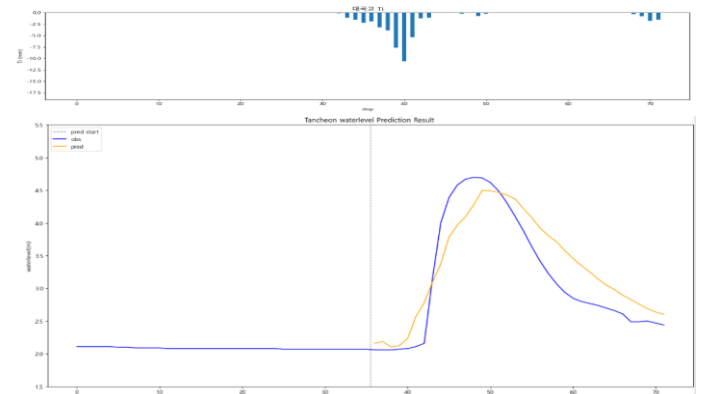
10분 후

R2 : 0.7413



40분 후

R2 : 0.8056

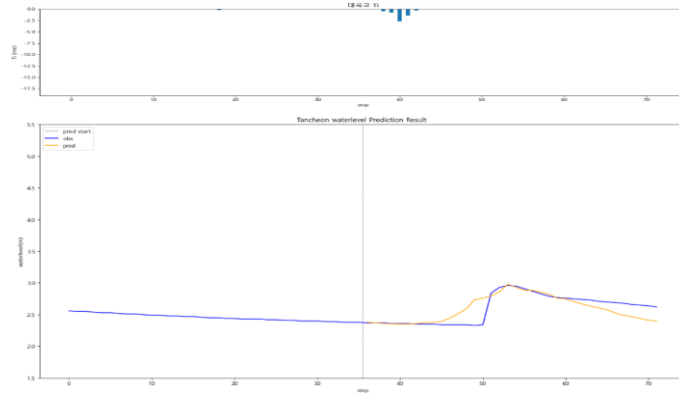


대곡교 - 10분 단위

* model_(gn/dg)_20

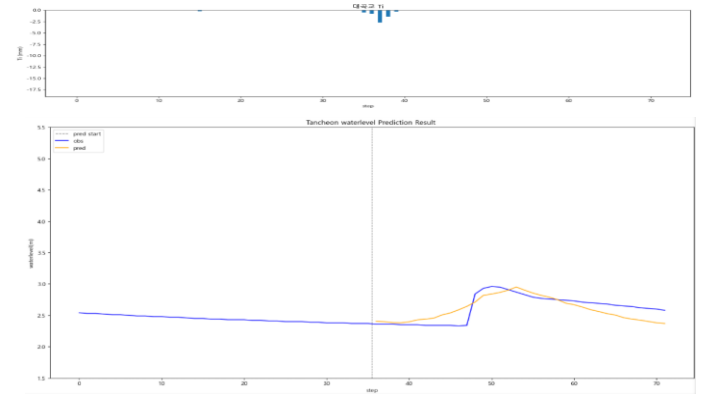
10분 전

R2 : 0.5765



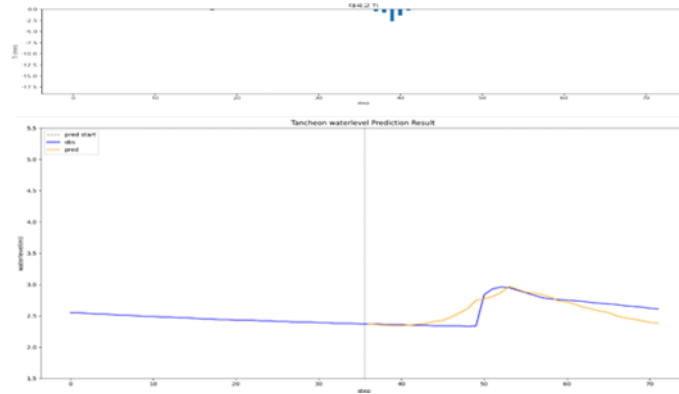
20분 후

R2 : 0.5821



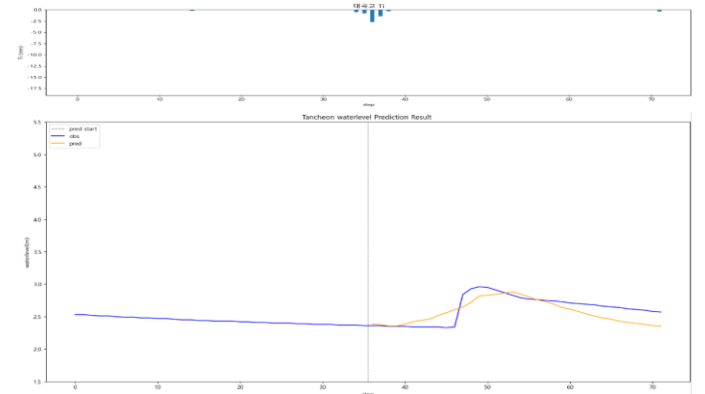
testdata_ 240723 (18)

R2 : 0.6077



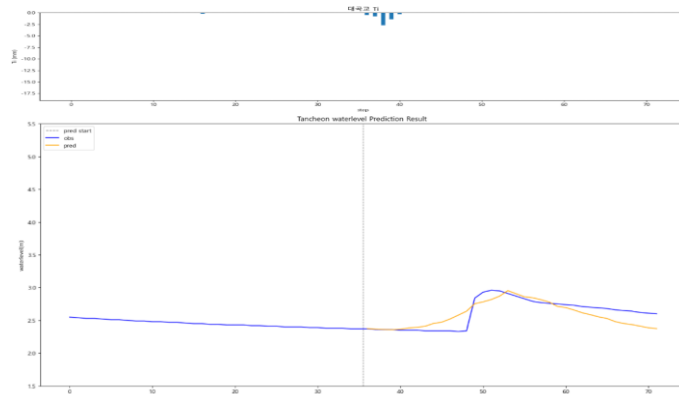
30분 후

R2 : 0.5114



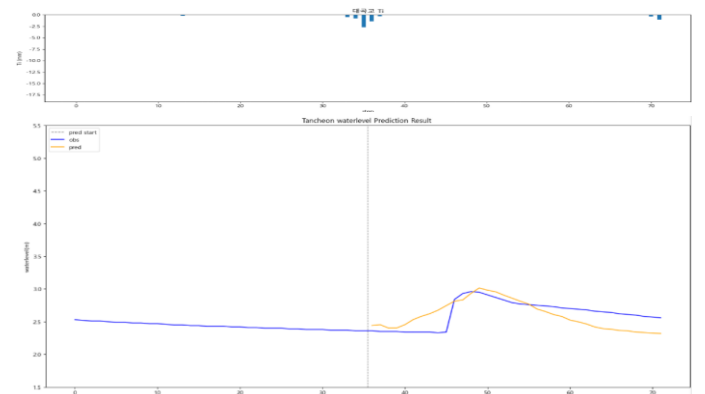
10분 후

R2 : 0.6268



40분 후

R2 : 0.1481

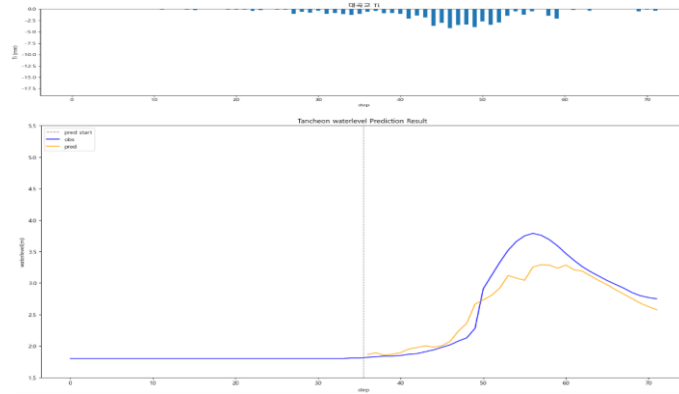


대곡교 - 10분 단위

* model_(gn/dg)_20

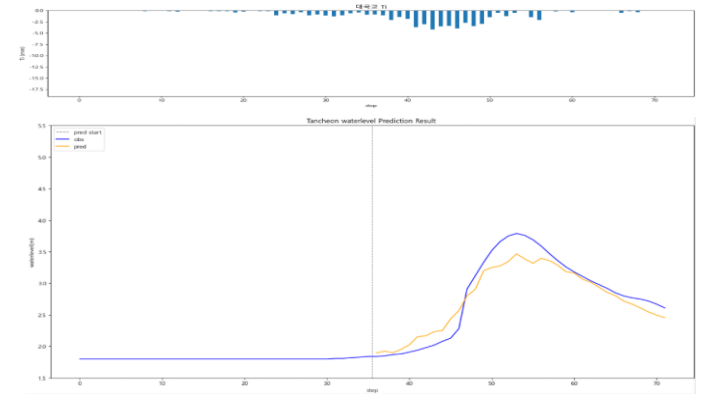
10분 전

R2 : 0.8560



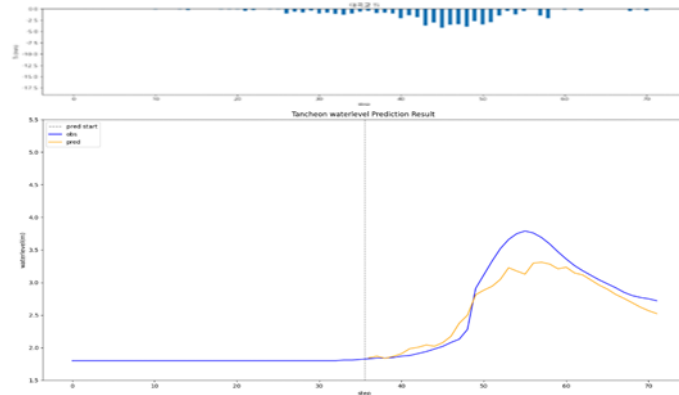
20분 후

R2 : 0.9062



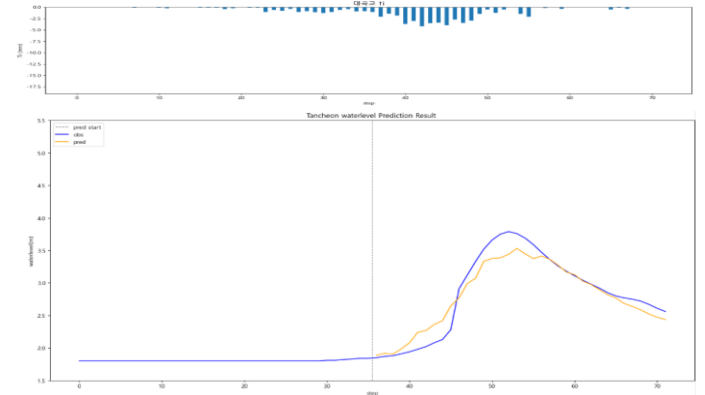
testdata_ 250716 (17)

R2 : 0.8647



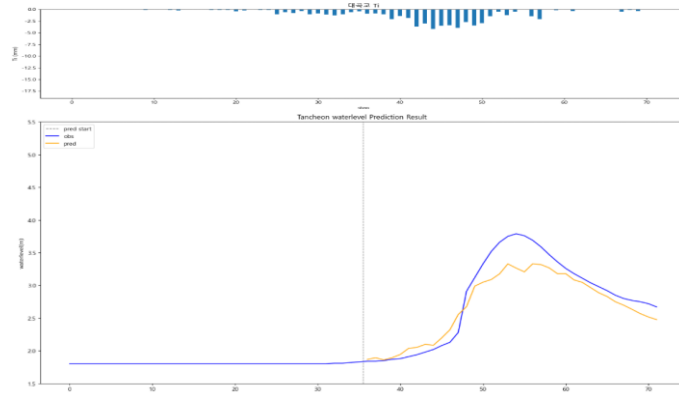
30분 후

R2 : 0.9148



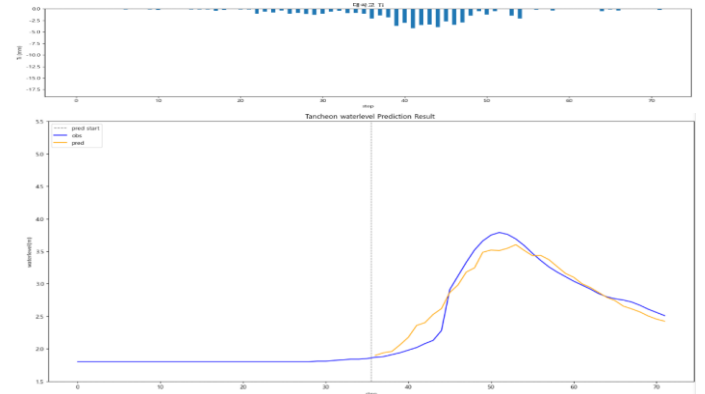
10분 후

R2 : 0.8740



40분 후

R2 : 0.9243

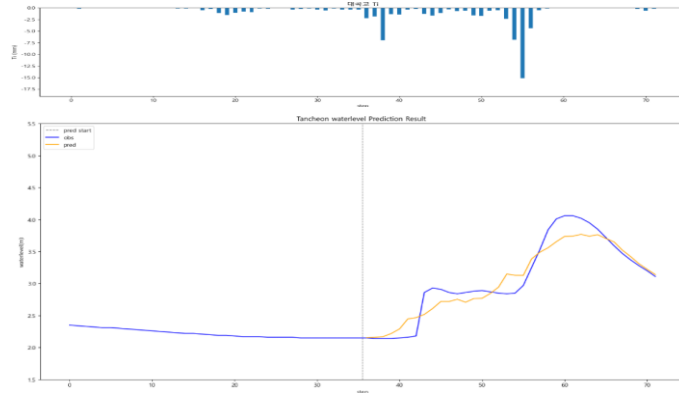


대곡교 - 10분 단위

* model_(gn/dg)_20

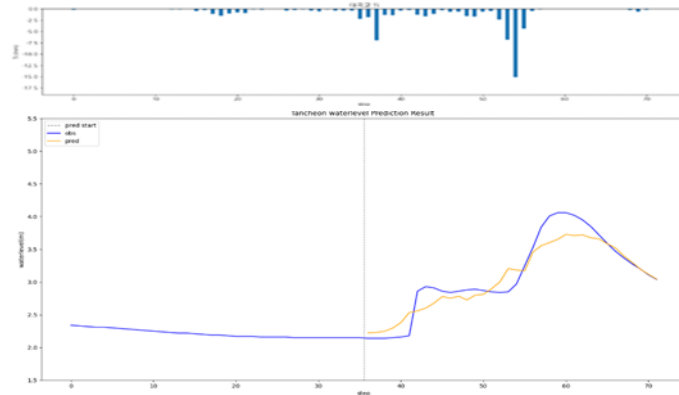
10분 전

R2 : 0.9033



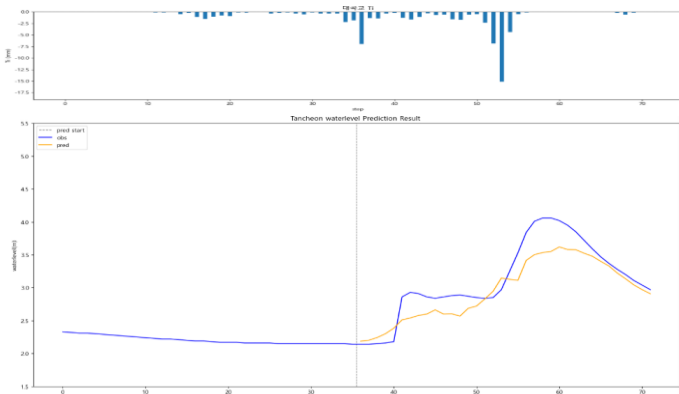
testdata_ 250814 (05)

R2 : 0.8840



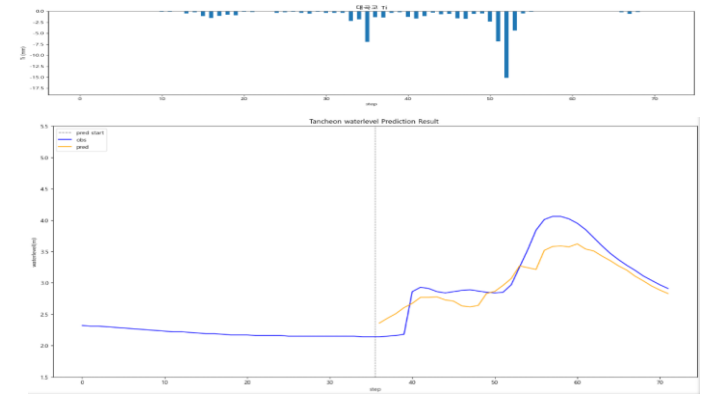
10분 후

R2 : 0.7832



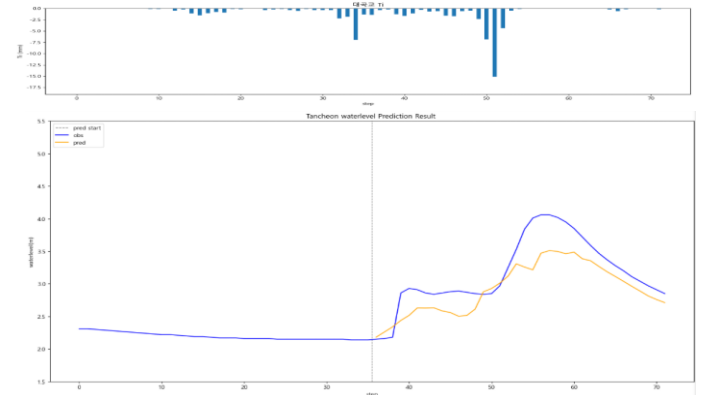
20분 후

R2 : 0.7671



30분 후

R2 : 0.6061



40분 후

R2 : 0.6646

